CSE563 — Unit IV Numerical Worksheet

Graph Theory

40 questions • CO4 • Student Worksheet

Instructions

Attempt all 40 questions. For every algorithm trace, **show the step-by-step table** — the table carries most of the marks, not the final answer. Break ties by choosing the alphabetically or numerically first vertex, and say so. Verify your answers where a check exists (max flow = cut capacity; MST has $n-1$ edges; BFS distance = tree-path length).

A. Graph Basics and Types (Q1–Q10)

Q1. A graph has degree sequence $(4, 4, 4, 3, 3, 2)$. Find the number of edges. Is such a graph possible?

Q2. Determine whether a simple graph can exist with degree sequence $(5, 4, 3, 2, 1, 1)$.

Q3. How many edges does the complete graph K_8 have, and what is the degree of each vertex?

Q4. How many edges does the complete bipartite graph $K_{4,6}$ have? Is it bipartite? Is it complete?

Q5. A tree has 15 vertices. How many edges does it have? How many edges must be added to make it contain exactly 3 cycles?

Q6. Write the adjacency matrix for the graph with $V = \{1, 2, 3, 4\}$ and $E = \{12, 13, 23, 34\}$, and compute the degree of each vertex from it.

Q7. For the path graph $1 - 2 - 3 - 4$, compute A^2 and state the number of walks of length 2 from vertex 1 to vertex 3.

Q8. Determine whether the graph with edges AB, BC, CD, DA (a 4-cycle) is Eulerian, and whether it is Hamiltonian.

Q9. A connected graph has degrees $3, 3, 2, 2, 2$. Does it have an Euler circuit, an Euler path, or neither?

Q10. Determine whether $K_{3,3}$ is planar, using Euler's inequality for bipartite planar graphs, $m \leq 2n - 4$.

B. Traversal: BFS, DFS and Topological Sort (Q11–Q18)

Q11. For the graph with adjacency lists 1:[2, 3], 2:[1, 4, 5], 3:[1, 5], 4:[2, 6], 5:[2, 3, 6], 6:[4, 5], perform BFS from vertex 1. Give the visit order and all distances.

Q12. For the same graph, perform DFS from vertex 1 (recursive, neighbours in listed order) and give the visit order and the tree edges.

Q13. Using the BFS result of Q11, give the BFS tree edges and verify that the tree path to vertex 6 has length equal to $d(6)$.

Q14. Use BFS to determine whether the graph of Q11 is bipartite.

Q15. Determine whether the 5-cycle $A - B - C - D - E - A$ is bipartite, justifying your answer two different ways.

Q16. Perform a topological sort of the DAG with edges $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$ using Kahn's algorithm.

Q17. Course prerequisites: Maths $\to$ Physics, Maths $\to$ Programming, Physics $\to$ Mechanics, Programming $\to$ DataStructures, DataStructures $\to$ Algorithms, Mechanics $\to$ Robotics, Algorithms $\to$ Robotics. Give a valid study order.

Q18. Explain, using a DFS trace, how a back edge proves the presence of a cycle in the graph with edges $A \to B$, $B \to C$, $C \to A$.

C. Dijkstra and Minimum Spanning Trees (Q19–Q28)

Q19. Run Dijkstra from S on the weighted graph with edges SA=1, SB=4, AB=2, AC=6, BC=3, CD=2, BD=7. Give the table and all final distances.

Q20. For the previous question, give the shortest path from S to D and its cost.

Q21. Explain why Dijkstra's algorithm gives the wrong answer on the graph $S \to A$ (weight 2), $S \to B$ (weight 6), $B \to A$ (weight -5). Give the correct distances.

Q22. Apply Kruskal's algorithm to the graph with edges AB=3, AC=1, BC=2, BD=5, CD=4, CE=6, DE=2. List accepted and rejected edges and give the total weight.

Q23. Apply Prim's algorithm starting from vertex A to the same graph, and confirm the total weight matches Kruskal's.

Q24. A network of 6 offices must be connected by cable. Costs (in lakhs): 12=4, 13=3, 23=1, 24=5, 34=6, 35=8, 45=2, 46=7, 56=9. Find the minimum total cost using Kruskal's algorithm.

Q25. For a graph where all edge weights are distinct, state whether the MST is unique, and justify.

Q26. Explain the difference between the MST and the shortest-path tree, giving a triangle as a counterexample.

Q27. Give the time complexity of BFS, DFS, Dijkstra (binary heap), Kruskal and Prim, in terms of $n = |V|$ and $m = |E|$.

Q28. A graph is stored as an adjacency matrix. State the cost of (a) checking whether edge (u, v) exists, (b) listing all neighbours of u, (c) the space used. Compare with adjacency lists.

D. Graph Colouring (Q29–Q33)

Q29. Find the chromatic number of the complete graph K_5, of a tree with 10 vertices, of the cycle C_6 and of the cycle C_7.

Q30. Apply Welsh–Powell to the graph with edges $AB, AC, AD, BC, BD, CE, DE$. Give the colour classes and $\chi(G)$.

Q31. Six examinations must be scheduled. Conflicts: E_1E_2, E_1E_3, E_2E_3, E_2E_4, E_3E_5, E_4E_5, E_4E_6, E_5E_6. Find the minimum number of slots and give a timetable.

Q32. State and justify the bounds $\omega(G) \le \chi(G) \le \Delta(G) + 1$ for a graph with maximum degree 5 containing a triangle.

Q33. Explain how graph colouring models register allocation in a compiler, and state what a colouring failure means in that context.

E. Network Flow (Q34–Q40)

Q34. Define a flow network, a valid flow, and the value of a flow. State the two constraints a flow must satisfy.

Q35. For the network $s \to a$ (10), $s \to b$ (5), $a \to b$ (4), $a \to t$ (7), $b \to t$ (10), find the maximum flow by Ford–Fulkerson, showing each augmenting path.

Q36. For the network of the previous question, identify the minimum cut and verify the max-flow min-cut theorem.

Q37. In the network $s \to a$ (6), $s \to b$ (4), $a \to c$ (3), $a \to d$ (4), $b \to d$ (5), $c \to t$ (5), $d \to t$ (6), find the maximum flow and the minimum cut.

Q38. Explain why reverse (residual) arcs are necessary in Ford–Fulkerson, using a concrete example where a greedy first choice would otherwise block the optimum.

Q39. Four students $S_1, \ldots, S_4$ must be assigned to four projects $P_1, \ldots, P_4$. S_1 can take P_1, P_3; S_2 can take P_1, P_2; S_3 can take P_2, P_3; S_4 can take P_4. Model this as a max-flow problem and find the maximum matching.

Q40. Repeat the previous question with S_1 able to take only P_1, S_2 only P_1, S_3 taking P_1, P_2, and S_4 taking P_4. Find the maximum matching and explain why a complete assignment is impossible.